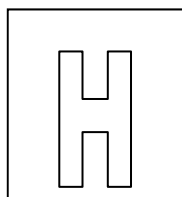


Candidate Name: _____

Class Adm No

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2017 Preliminary Examination 2 Pre-University 3

H2 Biology

9648/03

Applications Paper and Planning Question

19 September 2017

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.
You will lose marks if you do not show your working or if you do not use appropriate units.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
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5	
Total	

This question paper consists of 20 printed pages

[Turn over

Answer **all** questions.

- The ability to model human diseases using cultured pluripotent stem cells (PSCs) has revolutionized the ways in which we study monogenic, complex and epigenetic disorders, as well as early- and late-onset diseases. Several strategies are used to generate such disease models using either embryonic stem cells (ES cells) or patient-specific induced PSCs (iPSCs), creating new possibilities for the establishment of models and their use in drug screening. Fig. 1.1 shows strategies for generating human pluripotent stem cells (hPSCs) carrying a genetic disorder for research purposes. Disease-specific ES cells can be identified during the in-vitro fertilization (IVF) process by pre-implantation genetic diagnosis (PGD) or pre-implantation genetic screening (PGS).

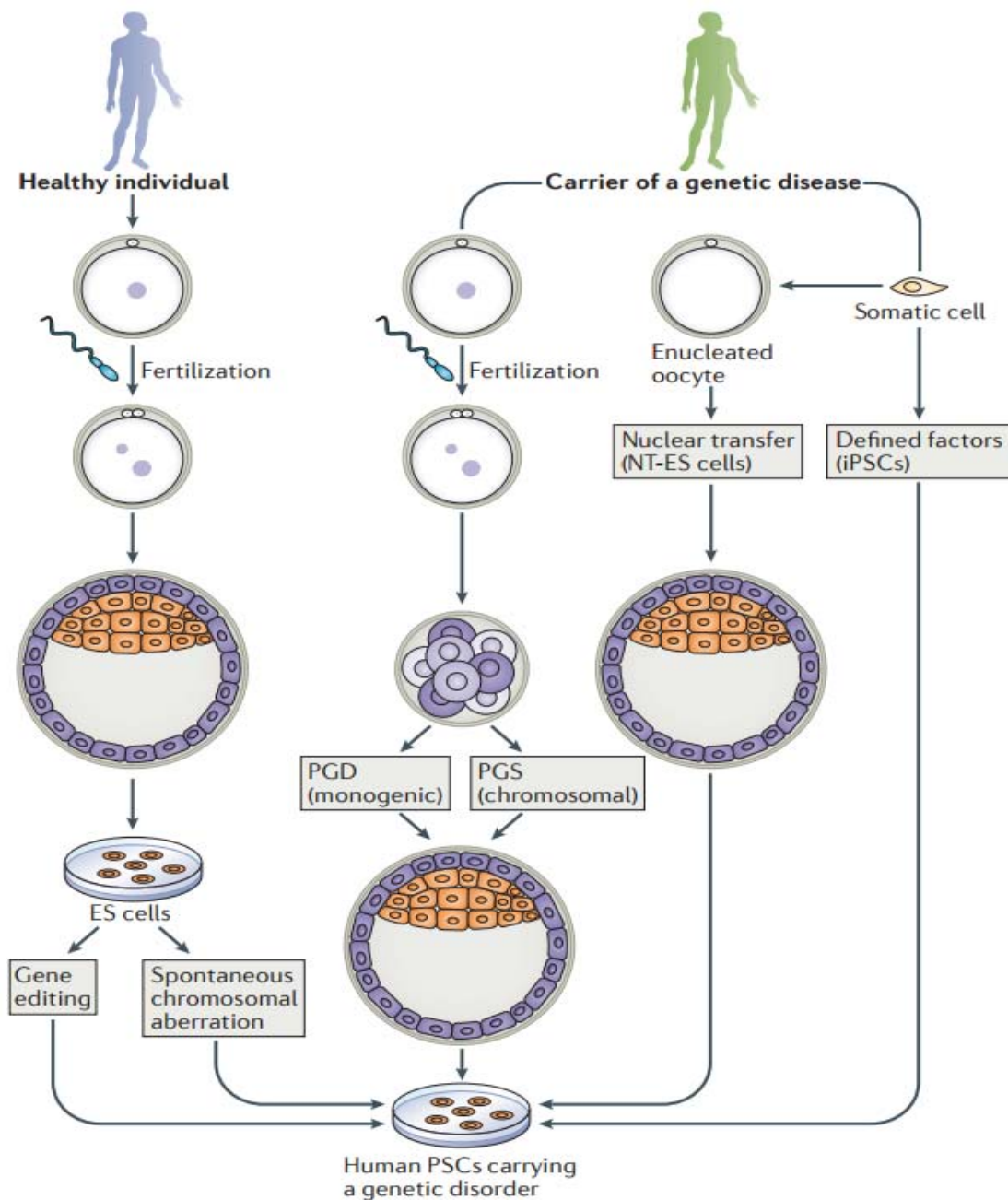


Fig. 1.1

(a) Describe one similarity and one difference between a blastocyst and embryo.

- Both blastocysts and embryos are a cluster of undifferentiated cells;
 - Both are pluripotent;
 - Both have high telomerase activity;
 - Both have self-renewal property that can undergo indefinite rounds of replication;
 - Blastocysts are earlier stage of embryo;
 - Blastocysts consist of inner cell mass but an embryo does not have an inner cell mass;
 - Blastocysts consist of cells that will form the placenta but embryo does not have those cells;
-[2]

(b) Describe one advantage and one limitation of using somatic cell nuclear transfer (SCNT) to generate human PSCs carrying a genetic disorder.

- Easier to isolate nucleus with defective allele from somatic cells;
 - Easier to harvest eggs and somatic cells;
 - Pluripotent stem cell is genetically identical to the patient;
 - Able to harvest pluripotent stem cells without the need to kill embryos;
 - Low success rate;
 - Cells obtained from SCNT may have additional abnormalities and chromosome aberrations due to incorrect reprogramming;
-[2]

In order to optimise the conditions and increase the chances of creating iPSCs from somatic cells, extensive research has been conducted on 4 main genes, *Oct4*, *Sox2*, *Nanog* and *Lin28* using M4 cell cultures. Fig. 1.2 shows the effect of different combinations of genes in the reprogramming mixture on the number of induced-pluripotent stem cell colonies formed.

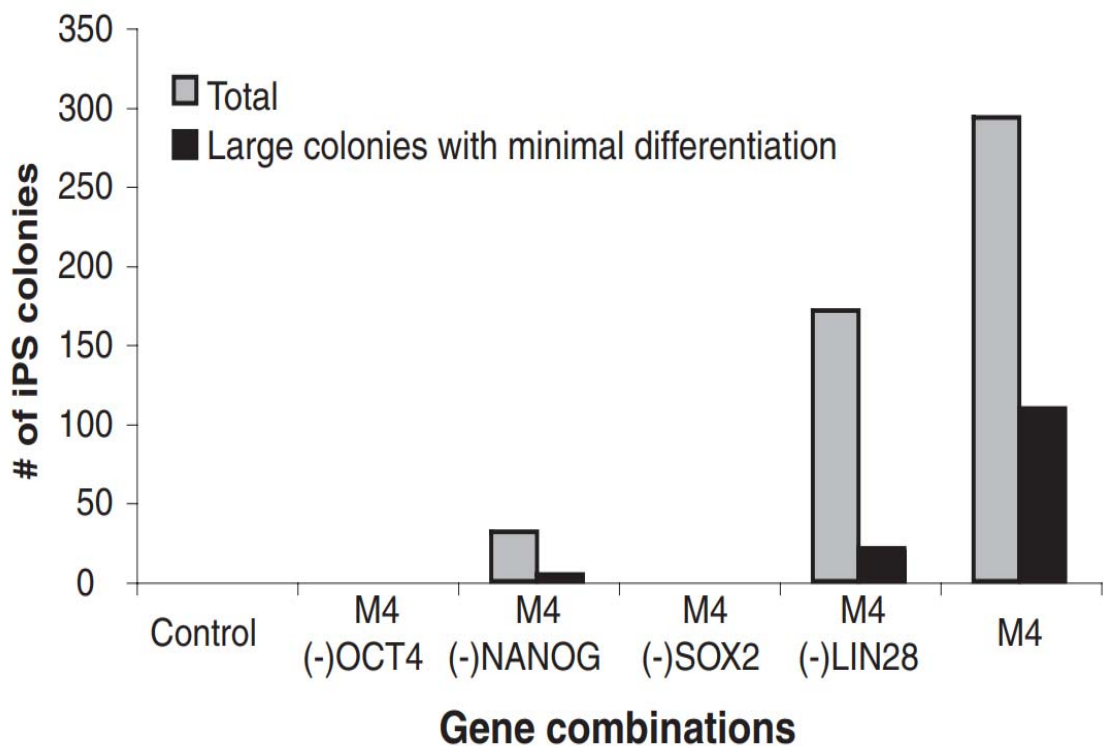


Fig. 1.2

- (c) With reference to Fig. 1.2,
(i) explain the purpose of the control.

The control group in this experiment are somatic cells that are not treated with any reprogramming genes to investigate the extent of each of the 4 genes that can reprogram somatic cells to form iPSCs;

.....
.....[1]

- (ii) describe the results produced from varying gene combinations in the reprogramming mixture.

- Oct4 and Sox2 are the more important genes with significant effect in reprogramming somatic cells to iPSCs as M4 without Oct4 and M4 without Sox2 shows zero colony of iPSCs.;
- Lin28 is the least significant gene in reprogramming somatic cells to iPSCs as 25 undifferentiated iPSC colonies / 180 total colonies can still be obtained even though M4 does not have Lin28;

.....
.....[2]

- (d) Describe one possible regulatory process at the chromosomal level that could increase the expression of *Oct4*, *Sox2*, *Nanog* and *Lin28* genes.

- Demethylation of DNA where methyl groups of certain DNA sequences are removed.
- Histone methyltransferase causes methylation of histone tails;
- allowing the DNA segment containing the 4 genes to be more accessible for RNA polymerase to bind to for transcription;
- Acetylation of histone tails lead to weaker binding between DNA and histones by Histone Acetylase (HAT);
- allowing the DNA segment containing the 4 genes to be more accessible for RNA polymerase to bind to for transcription;
- Demethylation of histone tails lead to weaker binding between DNA and histones by histone demethylase;
- allowing the DNA segment containing genes to be more accessible for RNA polymerase to bind to for transcription;

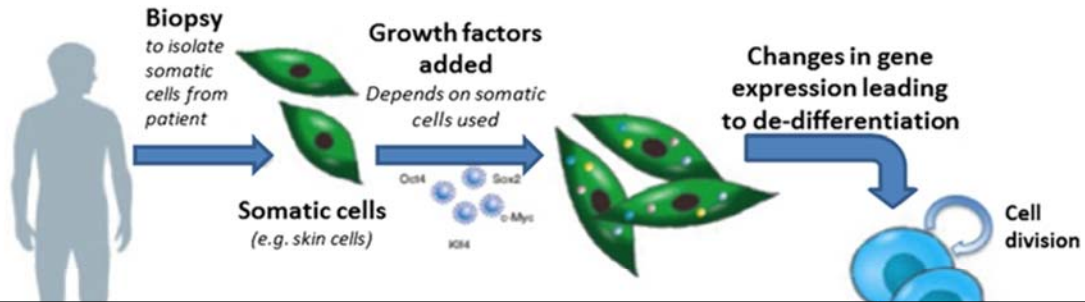
- (e) Suggest one reason why the number of cell colonies with minimal differentiation differs from the total number of cell colonies produced.

- There are other genes involved in reprogramming somatic cells into undifferentiated iPSCs that are not included in the mixture;
- Reprogramming of the nucleus is not always successful for all the somatic cells;

.....
.....[1]

X-linked Severe Combined Immunodeficiency (SCID) is a rare congenital disorder characterised by improper development of immune cells which has been treated by gene therapy. The ability to generate iPS cells that have similar characteristics with embryonic stem cells has provided a promising alternative to the use of haematopoietic stem cells for gene therapy to treat X-linked SCID.

Fig 1.3 shows a possible process of using iPS cells for gene therapy.



Use of liposome as vector

- Lower probability of liposomes binding to cell surface membrane of iPS cells and membrane fusion occurring to release DNA into target cells;
- therefore not all target cells receive normal IL2RG allele, some still express non-functional IL2RG protein;

OR

- Normal IL2RG allele not integrated into the iPS cell's DNA, unless retroviral vector is used;
- this leads to transient expression of normal functional proteins and multiple treatments are required;

OR

Gene therapy does not offer complete cure

- Difficult to control the expression of the normal IL2RG alleles
- Expression of proteins may be unstable – there may be too much or too little proteins being expressed → immune cells may have insufficient cytokine receptors/are abnormal, cannot carry functions;

(carried out using liposomes)

Fig. 1.3

- (f) Besides difficulties with de-differentiating somatic cells to iPS cells, explain one other factor that could prevent this method of gene therapy for X-linked SCID from becoming an effective treatment.

.....

.....

Against use of ES cells

- involves removal of inner cell mass from blastocyst → destruction of embryo which has the potential/ability to develop into a foetus/a human being → akin to murder/killing of a life for own benefit/to treat own disease;
- ES cells are able to divide continuously via mitosis → potential of developing tumours, causing more harm to the patient;
- ES cells could potentially develop into individuals and they did not give their consent for experimental use;

AND

- Disagree → iPS cells have characteristics of ES cells and hence have the potential of developing into a develop into a foetus/a human being;

OR

- Agree → iPS cells were not harvested from embryos but were obtained from de-differentiation of somatic cells → no destruction of human life so it is acceptable;
- Agree → Donors of IPS cells would be able to give their consent for experimental use;

.....[2]

[Total: 14]

2. Conventional DNA ladders are traditional molecular weight standards used for sizing and approximate quantification of linear double-stranded DNA fragments in agarose and non-denaturing polyacrylamide gels for research purposes. The markers are composed of lambda phage DNA digested to completion with the appropriate restriction enzyme(s), purified and dissolved in storage buffer. The DNA fragments contain blunt or sticky ends depending on the restriction enzyme used for the marker's preparation. Fig. 2.1 shows the genome of Lambda DNA with the restriction sites corresponding to six different restriction enzymes.

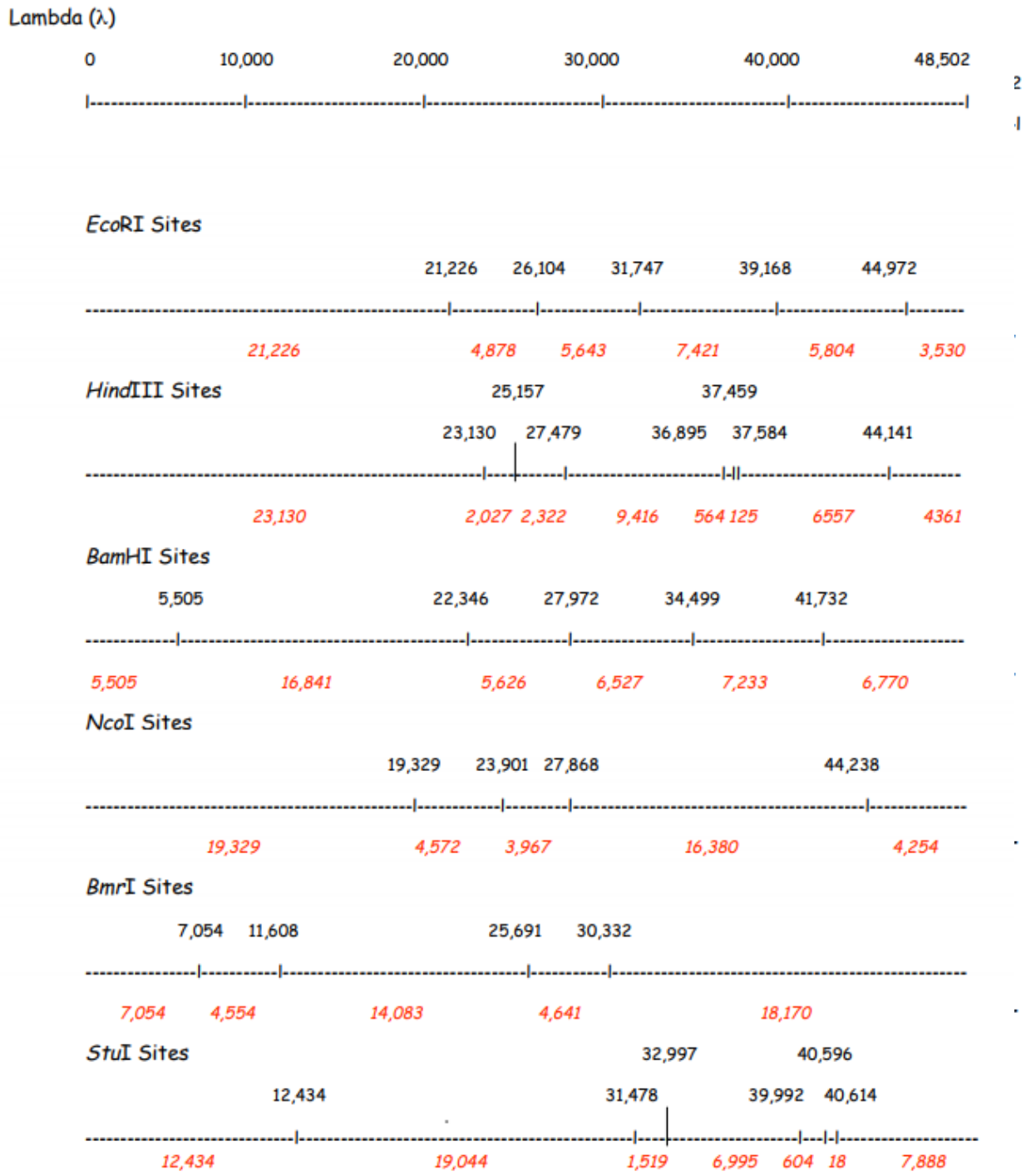


Fig. 2.1

Table 2.1

EcoRI	HindIII	BamHI	NcoI	Bmrl	StuI
21226	23130	16841	19329	18170	19044
7421	9416	7233	16370	14083	12434
5804	6557	6770	4572	7054	7888
5643	4361	6527	4264	4641	6995
4878	2322	5626	3967	4554	1519
3530	2027	5505			604
	564				18
	125				

(a) With reference to Fig. 2.1, fill in the columns of Table 2.1 with the respective DNA fragments generated from their corresponding restriction enzymes. List each fragment from the largest to the smallest. [2]

(b) Explain two factors that would influence a researcher's decision in choosing a restriction enzyme for the restriction digest step of the gene cloning experiment.

- The position of the restriction site of the restriction enzyme as the researcher would not choose a restriction enzyme that can target a sequence within the gene of interest as a restriction site;
- Proximity of restriction site to the gene of interest as the researcher would choose a restriction enzyme that can excise the gene at restriction sites that flank right next to the gene;
- Restriction enzyme must be able to target restriction sites on both the vector and gene of interest;
- Restriction enzyme would be preferred if it generates sticky ends because if it generates blunt ends, it requires additional steps of adding linker DNA;
- Restriction enzyme chosen should make a single and unique cut to the vector as the researcher would not want to fragment the vector / gene of interest is not incorporated into multiple sites;
- Availability of the restriction enzyme;
- Cost of the restriction enzyme;
- AVP;

[2]

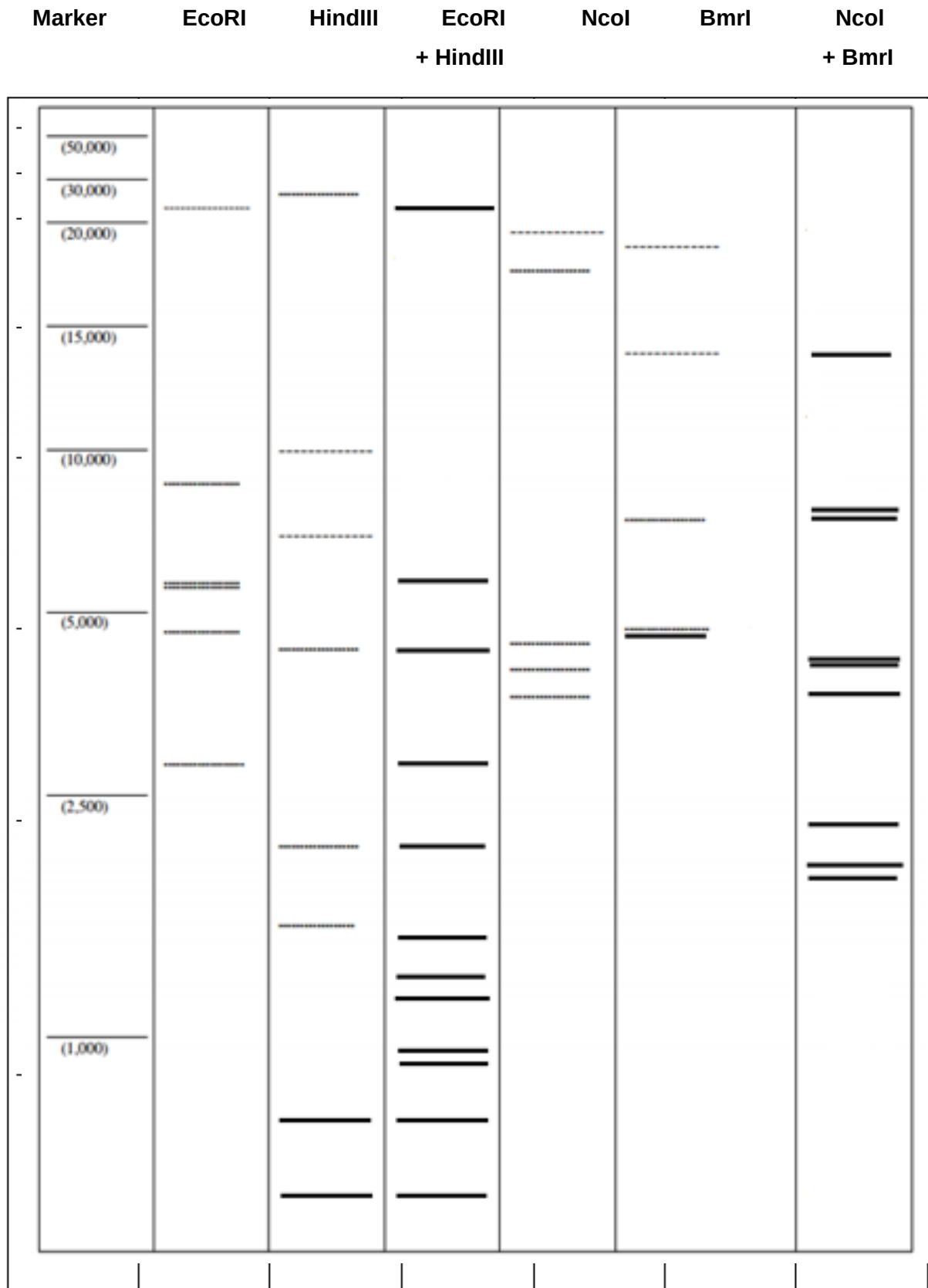


Fig. 2.2

- (c) Complete Fig. 2.2 below by drawing the DNA band patterns after gel electrophoresis. [2]

(d) Explain how DNA bands can be visualized after gel electrophoresis.

- o Ethidium bromide / Bromophenol blue / xylene cyanol is added to the gel in order to facilitate visualization of DNA after electrophoresis;
- o Ethidium bromide / Bromophenol blue / xylene cyanol binds to only DNA;
- o DNA bands are viewed under UV light;

.....[2]

(e) Suggest a method to improve the separation of DNA bands during gel electrophoresis.

- Decrease voltage during gel electrophoresis;
- Increase the concentration of agarose used to cast the gel;
- Add glycerol to the DNA samples in the loading well;
- Cast an agarose gel that is longer in length;
- AVP;
- R! Increase duration of electrophoresis

.....[1]

Besides being used for DNA fragment separation, gel electrophoresis can also be conducted to separate protein fragments for agricultural research purposes in order to study relationships between protein band patterns and phenotypic traits.

Barley (*Hordeum vulgare*) is an important crop in southern Brazil where its production is used in the brewing industry. Hence, the malting quality of different barley plant cultivars must be continuously researched and improved upon. Cultivars are new plant species obtained via artificial selecting breeding processes.

Malt is germinated cereal grains that have been dried via malting. Malting grains develop the enzymes required for modifying the starch in the grains into various types of sugars such as maltose and glucose. Characteristics of importance for malting quality, which can differ considerably among barley cultivars, include grain size, grain protein concentration and nitrogen content in the seeds.

Recently, researchers are researching on how the quality of a particular storage protein named hordein could affect the malting quality of barley plants. Barley is highly polymorphic regarding the hordein polypeptide composition. Electrophoresis is particularly attractive as a screening test to differentiate barley plant cultivars and to determine malting quality of each variety. By comparing the total hordein pattern from barley cultivars of different malting quality, researchers can investigate the relationship between malting quality and band patterns and to explore the feasibility of using hordein protein electrophoresis to assist in the selection of barley plant cultivars for malting.

Fig. 2.3 shows the hordein polypeptide band patterns of 14 different barley plant species. On the right of the gel, a polypeptide fragment ladder showing the band positions of 26 different hordein polypeptide sizes serves as a reference point for comparison.

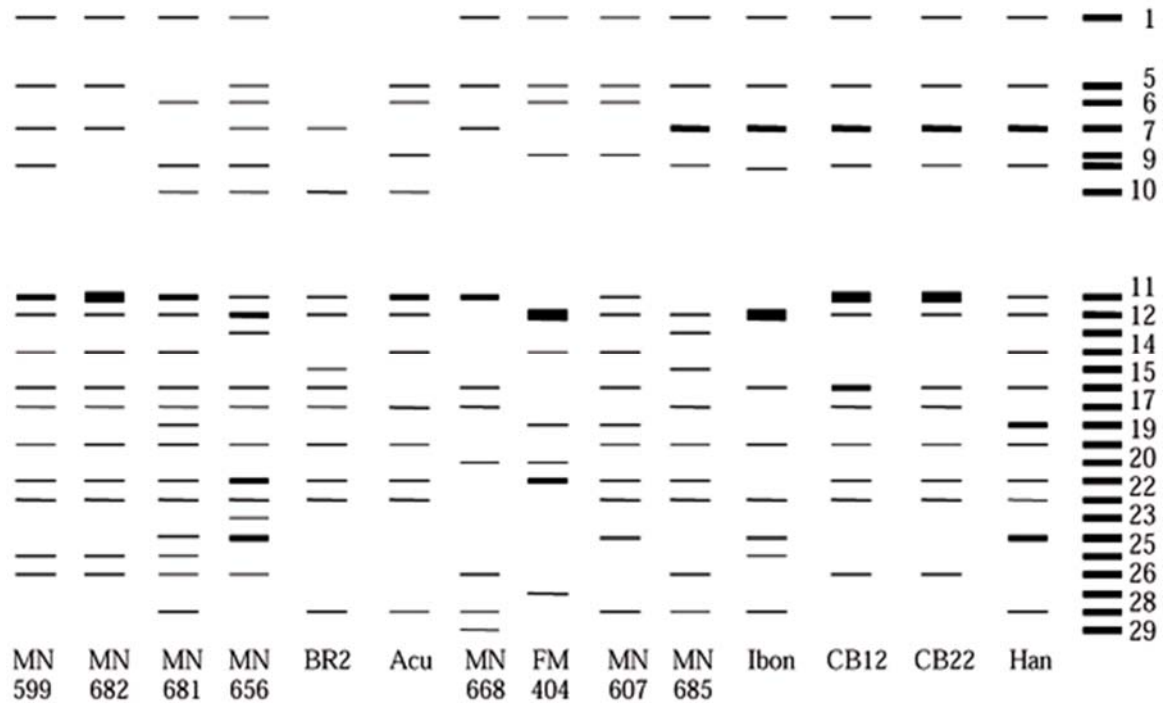


Fig. 2.3

(f) With reference to Fig. 2.3, complete the phylogenetic tree shown in Fig. 2.4.

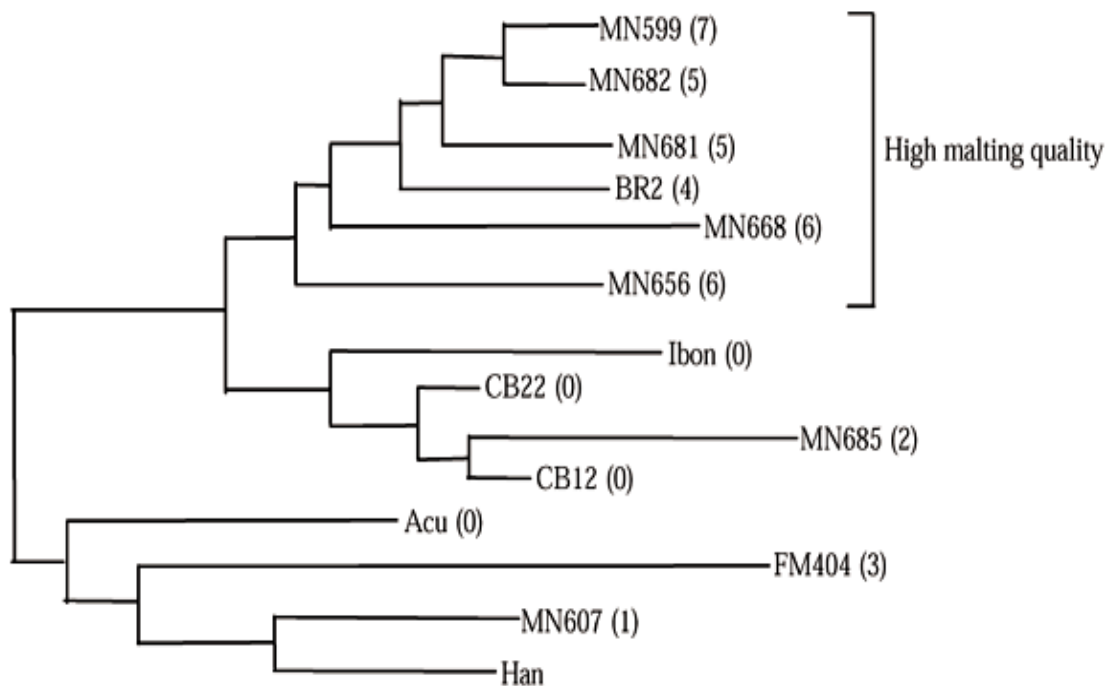


Fig. 2.4

[2]

Table 2.2 shows the correlation of each polypeptide band with the malting quality of barley varieties studied. The number of + / - in Table 2.2 indicates the strength of correlation between the polypeptide band and malting quality.

Table 2.2

Band	Correlation	Band	Correlation
1	-	17	-
5	--	18	-
6	+	19	---
7	-	20	+
8	-	21	-
9	-	22	-
10	+	23	+
11	+	24	-
12	--	25	+
13	+	26	+
14	-	27	+
15	+	28	-
16	+	29	+

Table 2.3 shows the frequency of the hordein polypeptide bands (in percentage) in each barley variety studied.

Table 2.3

Hordein band	Varieties ⁽¹⁾													
	MN	MN	MN	MN	MN	BR2	Acu	Han	FM	MN	MN	Ibon	CB12	CB22
	599	682	668	681	656				404	607	685			
1	20	20	27	27	38	0	0	78	100	70	100	90	100	100
5	20	10	18	0	8	0	91	71	100	70	100	90	100	90
6	0	0	0	9	8	0	75	0	100	70	0	0	0	0
7	20	20	27	0	31	13	0	78	0	0	100	90	100	100
8	0	0	0	0	0	0	75	0	100	50	0	0	0	0
9	20	0	0	27	8	0	0	64	0	0	100	70	100	90
10	0	0	0	9	8	33	8	0	0	0	0	0	0	0
11	100	100	100	100	54	100	100	100	0	100	0	0	100	100
12	100	70	0	64	92	40	100	100	100	100	100	100	100	100
13	0	0	0	0	46	0	0	0	0	0	100	0	0	0
14	30	30	0	27	0	0	100	100	100	100	0	0	0	0
15	0	0	0	0	0	60	0	0	0	0	100	0	0	0
16	100	100	100	100	54	100	0	100	0	30	0	100	100	100
17	50	60	100	73	23	60	91	0	0	0	100	0	100	100
18	0	0	0	18	0	0	0	100	100	100	0	0	0	0
19	20	20	0	36	54	66	100	100	0	90	100	100	100	100
20	0	0	9	0	0	0	0	0	100	0	0	0	0	0
21	100	100	0	100	46	100	42	100	100	100	100	0	100	100
22	100	100	0	100	46	100	91	36	0	100	100	100	100	100
23	0	0	0	0	23	0	0	0	0	0	0	0	0	0
24	0	0	0	27	54	0	0	100	0	100	0	90	0	0
25	50	30	0	45	0	0	0	0	0	0	0	20	0	0
26	40	40	100	9	23	0	0	0	0	0	100	0	100	100
27	0	0	0	0	0	0	0	0	100	0	0	0	0	0
28	0	0	64	9	0	6	8	93	0	100	50	100	0	0
29	0	0	27	0	0	0	0	0	0	0	0	0	0	0

- (g) With reference to Fig. 2.4, Table 2.2 and Table 2.3, identify and explain which barley variant would have the best malting quality.

- MN668;
- The presence of the hordein bands 5, 12 and 19 can be useful as molecular indicator of a low malting quality / Bands 5 and 12 and 19 strongly correlated negatively with malting quality scores;
- According to the phylogenetic tree diagram, MN688 also belongs to the family of barley plant variants of high malting quality;
- Lowest frequency of polypeptides band 12 and 19 at 0% and relatively low frequency of polypeptide band 5 at 18%;
- Presence of hordein band 29 is positively correlated to malting quality and MN668 has the highest frequency of polypeptides band 29 at 27% compared to the other variants;
- Bands 11, 16 and 19 correlate positively with malting quality scores and MN668 has 100% frequency of those 3 bands;

.....

[4]

[Total: 15]

3. Rice is the staple diet in many parts of the world. It lacks a number of important nutrients, including β carotene, from which vitamin A is synthesised. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease. Golden rice, which contains β carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium.

Fig. 3.1 shows an artificial DNA sequence used.



Key:

pro – promoter sequence for polymerase enzymes

ter – termination signal for polymerase enzymes

Hyg resist – Hygromycin B antibiotic resistance gene from a bacterium

Fig. 3.1

Fig. 3.2 shows the main events involved in obtaining a transgenic rice plant.

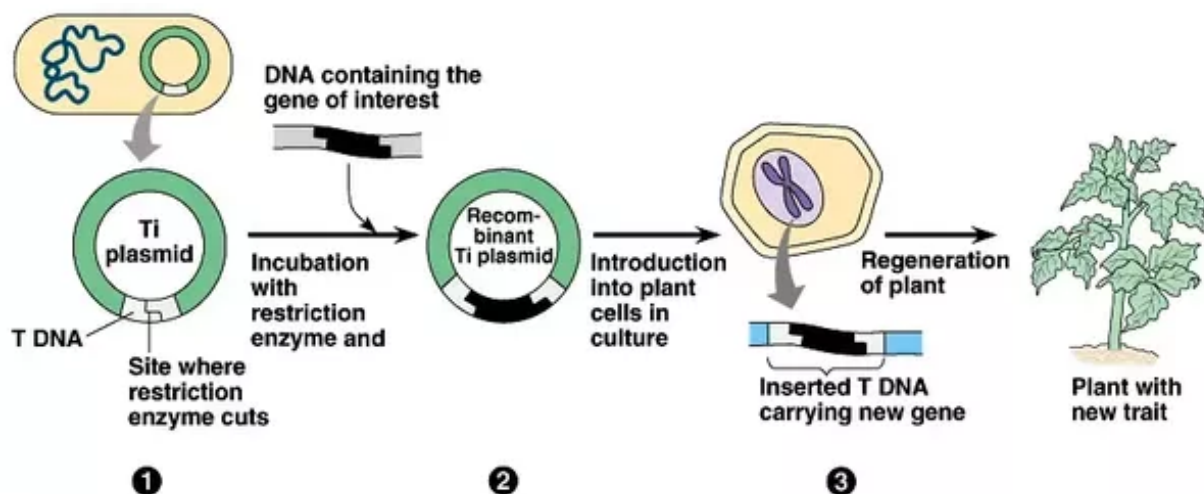


Fig. 3.2

- (a) Define the term 'transgenic'.

Relating to or denoting an organism that contains genetic material into which DNA from an unrelated organism has been artificially introduced.

...

.....[1]

(b) Describe two unique and distinct regions found within the Ti plasmid.

- Genes in the virulence region code for the enzymes responsible for mediating conjugative transfer of T-DNA to plant cells.
- T-DNA region consists of genes such as auxin, cytokinin and opine that code for plant growth regulators that alter the development and metabolism of the host plant cell;
- Ori which represents origin of replication where helicase would bind to to unwind DNA for DNA replication;
- Ti plasmids contain selectable marker allows for identification of transformed cells;
- Ti plasmids have multiple restriction sites, to allow introduction of genes;

.....[2]

(c) With reference to Fig. 3.1 and Fig. 3.2,

(iii) outline the processes involved from stage 2 to stage 3.

- Bacterial transformation via calcium chloride treatment + heat shock / electroporation to facilitate uptake of recombinant Ti plasmid into agrobacterium;
- Hyg resistance gene is also inserted into the recombinant Ti plasmid to enable selection of the transgenic cells such that only the bacterial cells with the new DNA can grow in the presence of Hygromycin B;
- The modified agrobacterium is placed in a liquid suspension to the leaves of susceptible plants, infecting them;
- The T-DNA is excised from Ti plasmid, transferred into nucleus and integrated into the plant genome;

.....[4]

(iv) outline the processes involved from stage 3 to generate a full-grown transgenic plant.

- The explant is then aseptically transferred to culture vessels containing aseptic culture medium that contains nutrients and plant growth regulators;
- Auxin and cytokinin in equal ratio is added to the nutrient agar to stimulate the cells of the explant to divide by mitosis to form a callus;
- As callus increases in size, it can be subcultured into many calli by placing them in separate culture vessels containing culture medium with plant growth regulators;
- The cells of a callus can be induced to either proliferate or differentiate into particular tissues
- Low level of cytokinin and high level of auxin triggers formation of roots growth / High level of cytokinin and low level of auxin triggers shoot growth / Auxin and gibberellin stimulates cell differentiation; (any one e.g.)
- With differentiation of cells of callus into particular tissues, a plantlet can form;
- Plantlets are then allowed to acclimatized before transfer to soil to grow into a whole plant;

.....[4]

[Total: 11]

4. A pineapple plantation owner wants to find out the amount of ascorbic acid (vitamin C) in the pineapples cultivated in the plantation. He believes that his pineapples produce the most vitamin C compared to the standard pineapple breeds which typically contain 40.0 – 48.5 %.

The amount of ascorbic acid present in a sample can be determined using the dye dichlorophenol indophenol (DCPIP). At pH 7 and above, ascorbic acid reduces solutions of the dye dichlorophenol indophenol (DCPIP) from blue to colourless. The pH of the samples must be adjusted to pH 9 for this experiment to work. Ascorbic acid does not chemically change when neutralised by sodium hydroxide or when boiled.

Using this information and your own knowledge, design an experiment to determine the validity of the plantation owner's claim that the pineapples from his plantation contain higher concentrations of ascorbic acid.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- 100 cm³ of 5.0 % stock solution of ascorbic acid, adjusted to pH 7
- 100 cm³ distilled water
- 100 cm³ molten agar containing DCPIP
- Sterile petri dishes
- 1ml syringes
- Plastic straw to create wells in the agar plate
- Labels
- Stopwatch
- Forceps
- Normal laboratory glassware e.g. test tubes, beakers, graduated pipettes, droppers, glass rods etc
- 10 cm³ pineapple juice, supplied by the plantation owner
- Sodium hydroxide
- pH meter

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variable,
- describe the method with the scientific reasoning used to decide the method so that results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of tables and graphs,
- use correct technical and scientific terms,
- include references to safety measures to minimize any risk associated with the proposed experiment.

[Total: 12]

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Introduction

- Ascorbic acid reduces blue DCPIP to colourless. .
- Increase in concentration of ascorbic acid will increase the rate of decolourisation of DCPIP
- Different concentrations of ascorbic acid can be created from the stock solution. A standard curve of the amount of decolourisation of DCPIP by the different concentrations of ascorbic acid can be created. The amount of ascorbic acid in pineapple can be determined by reading off the standard curve

Explain how to determine concentration of ascorbic acid in pineapples using the standard curve [1m]

Procedure

1. Obtain 10cm³ of different concentrations of ascorbic acid solution by dilution.

Concentration of ascorbic acid solution / mmolL ⁻¹	Volume of 5.0 mmolL ⁻¹ ascorbic acid solution / cm ³	Volume of distilled water / cm ³
5	10	0
4	8	2
3	6	4
2	4	6
1	2	8
0	0	10

Describe how to obtain different concentrations of ascorbic acid [1m]

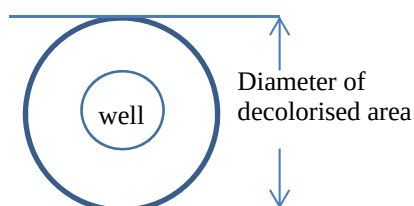
1. Pour the molten agar containing DCPIP into the petri dishes and allow the agar to cool.
2. Once the agar is cooled, use the plastic straw to make eight equal sized wells in the agar gel plate. Ensure that the wells are well-spaced.
3. Prepare a control experiment using boiled and cooled pineapple juice, following the same experimental procedures and conditions, to show that the decolourisation of DCPIP is due to the action of ascorbic acid and not due to the action of any enzymes in the juice.
4. Add 10% sodium hydroxide solution to the boiled and cooled pineapple juice, drop by drop with a dropper, until the pH is between 7 to 9. Check the pH by removing a drop of solution with a clean glass rod and placing it on indicator paper.
5. Neutralise the fresh pineapple juice in the same manner as described in step 4.
6. Using the 1 ml syringe, place 0.2 ml of each of the ascorbic acid solutions prepared according to the dilution table, 0.2 ml of pineapple juice and 0.2 ml of boiled and neutralised pineapple juice into one well each. Label the wells.
7. Replace the lid of the petri dish and leave the plates on the table for one hour.
8. After one hour, place the dish on the graph paper and measure the diameter of each of the rings where the blue DCPIP has been decolourised
9. Repeat step 1 to 7 three times.

Describe the settling up of the DCPIP agar plates and wells in the plates[1m]

Describe control to prove reaction is due to ascorbic acid in the pineapple juice and is not enzyme catalysed [1m]

Describe neutralisation of fresh pineapple juice.[1m]

State appropriate volumes scorbic acid [1m]



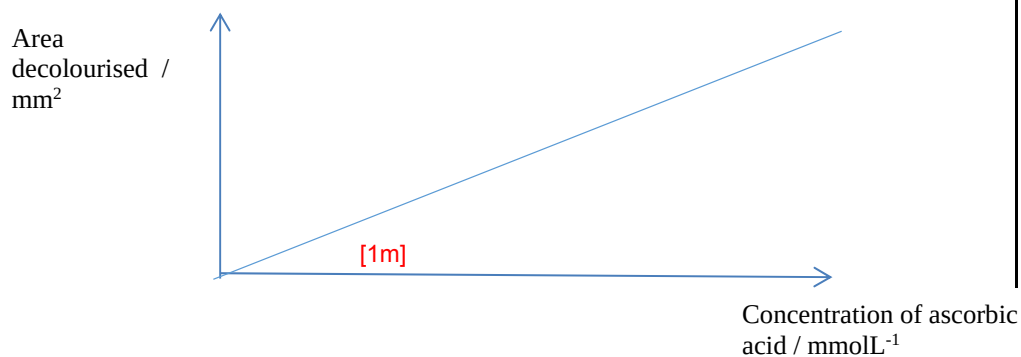
Describe the measurement of ring of decolourisation [1m]
Describe repeats [1m]

Draw a labelled diagram [1m]

10. Record the results in the table below and calculate the area of decolorisation

Concentration of ascorbic acid	Diameter of ring of decolourisation / mm				Area decolourised / mm ²
	Experiment 1	Experiment 2	Experiment 3	Average	
5					
4					
3					
2					
1					
0					
Sample of neutralised pineapple juice					
Sample of boiled and cooled pineapple juice which has been neutralised					

[1m]



From the standard curve drawn, calculate the concentration of ascorbic acid found in the sample of pineapple juice from the area of decolorisation obtained from the experiment with the sample of pineapple juice. If the concentration is higher than 0.8 to 1.6 mmolL⁻¹, the plantation owner's claim of his breed producing a higher concentration of Vitamin C than standard pineapple breeds is valid. [1m]

Safety

Sodium hydroxide and ascorbic acid may cause irritation when in contact with skin. Wear gloves when handling these reagents. [1m]

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5.

(a) Describe the use of restriction fragment length polymorphism analysis in creating a linkage map. **[6]**

- *Using RFLP with known physical location on a chromosome;*
- *Construct map using recombination frequency to determine order/sequence of genes and relative distance between different genes and genetic markers along a chromosome;*

; for 1 mark, max is 1 mark

- *A linkage map shows the relative location or the order of genes along a chromosome;*
- *constructed based on the assumption that the probability of a crossover between two genetic loci/RFLP markers is proportional to the distance separating the loci;*
- *Linkage maps are usually constructed with several thousand known genetic markers spaced evenly throughout the genome;*
- *The RFLP markers can be any genes or any other identifiable DNA sequences, such as variation number tandem repeats (VNTR) and short tandem repeats (STR);*

; for 1 mark, max is 2 marks

How to carry out linkage mapping using RFLP:

- *Linkage map must be done based on experimental crosses;*
- *After mating/fertilization, RFLP from parental and offspring organisms are obtained and analysed;*
- *Using gel E & Southern blot/nucleic acid hybridisation;*
- *RFLP pattern obtained are used to calculate the total percentage of recombinant offspring;*
- *Give an indication of the distance between the RFLP markers/sequences based on the recombination frequencies obtained;*
- *the farther apart the two RFLP markers/sequences are, the higher the probability that a crossover that generates genetic variation will occur between them and therefore the higher the recombination frequency;*
- *For example, if 70% of the progeny produced are parental and 30% were recombinant, the two RFLP loci are 30 centi-Morgans (cM) apart from each other;*

; for 1 mark, max is 3 marks.

- (b) Describe the processes involved in PCR.
[6]

Stage 1: Denaturation

- a) DNA denaturation by heating to 90°C; [accept 90 – 100°C] in a thermocycler;
- b) DNA is separated into single strands by breaking of hydrogen bonds;

Stage 2: Annealing of primers

- c) Annealing of primers to gene of interest by cooling to 54°C; [accept 30 – 65°C]
- d) primers/ oligonucleotides bind to single (DNA) strands / 3' ends;
- e) by annealing/hybridise to their complementary sequences on either side of the target sequence;

Stage 3: Extension

- f) DNA synthesis by heating to 72°C; [accept 60 – 75°C]
- g) optimum temperature (for Taq polymerase);
- h) new strands (of DNA) are synthesised by Taq/ DNA polymerase;
- i) starts at position of DNA primers;
- j) addition of free deoxyribonucleotide to the free 3'OH end;
- k) through the formation of phosphodiester bond between the nucleotides;
- l) using the single strand DNA (target sequence) as a template;

- (c) Discuss the goals and benefits of the Human Genome Project.
[8]

Goals

1. Construct a detailed genetic map (i.e. map formed using recombination frequencies and measured in terms of cM) of the entire human genome.;
2. Determine the nucleotide sequences of all 24 human chromosomes (i.e. the physical map of the genome as measured in base pairs) by the year 2005.;
3. Identify all the approximately 20,000-25,000 genes in human DNA.;
4. Improve technology for DNA sequencing and studying the function of DNA on a genomic scale.;
5. Sequence genomes of model organisms (*E. coli*, budding yeast, *C. elegans*, *Drosophila*, and mouse) in order to compare genomes.;
6. Develop bioinformatics support – to (a) create and operate databases for easy access to data and (b) develop and improve tools for data analysis eg. Comparing and interpreting genome information.;
7. Address the ethical, legal and social issues that may arise from the project.;

Max 4 marks

Benefits

A. Molecular medicine (no marks for heading; max 2 mks, @ 1 mk)

- 1 Earlier diagnosis/detection of genetic diseases;
- 2 Gene therapy;
- 3 Rational drug design/control systems for drugs/rational drug design/pharmacogenomics & custom drugs;

B. Energy and Environmental Applications (max 1 mk, @ 1 mk)

- 4 Use microbial genomics research to create new energy sources (biofuels);
- 5 Use microbial genomics research to develop environmental monitoring techniques to detect pollutants ;
- 6 Use microbial genomics research for safe, efficient environmental remediation;

C. DNA Forensics (max 3 mk, @ 1 mk)

- 7 Identify potential suspects whose DNA may match evidence left at crime scenes;
- 8 Exonerate persons wrongly accused of crimes;
- 9 Identify crime and catastrophe victims;
- 10 Establish paternity and other family relationships;
- 11 Identify endangered and protected species as an aid to wildlife officials (could be used for prosecuting poachers);
- 12 Detect bacteria and other organisms that may pollute air, water, soil, and food;
- 13 Match organ donors with recipients in transplant programs;
- 14 Determine pedigree for seed or livestock breeds;
- 15 Authenticate consumables such as caviar and wine;

D. Agriculture, Livestock Breeding, and Bioprocessing (max 1 mk, @ 1 mk)

- 16 Healthier, more productive, disease-resistant crops/ farm animals / higher yield;
- 17 More nutritious produce ;
- 18 Edible vaccines incorporated into food products;
- 19 New environmental cleanup uses for plants like tobacco;

E. Bioarchaeology, anthropology, evolution and human migration (max 1 mk, @ 1 mk)

- 20 Study human evolution (through germline mutations in lineages);
- 21 Study of migration of diff pop groups based on female genetic inheritance/lineage and migration of males via Y chromosomes;
- 22 Compare breakpoints in the evolution of mutations with ages of populations and historical events;

F. Risk assessment (@ 1 mk)

- 23 Assess health damage and risks caused by radiation exposure/mutagenic chemicals/ cancer-causing toxins;

Max 4 marks

[Total: 20]

End of Paper

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[Turn over